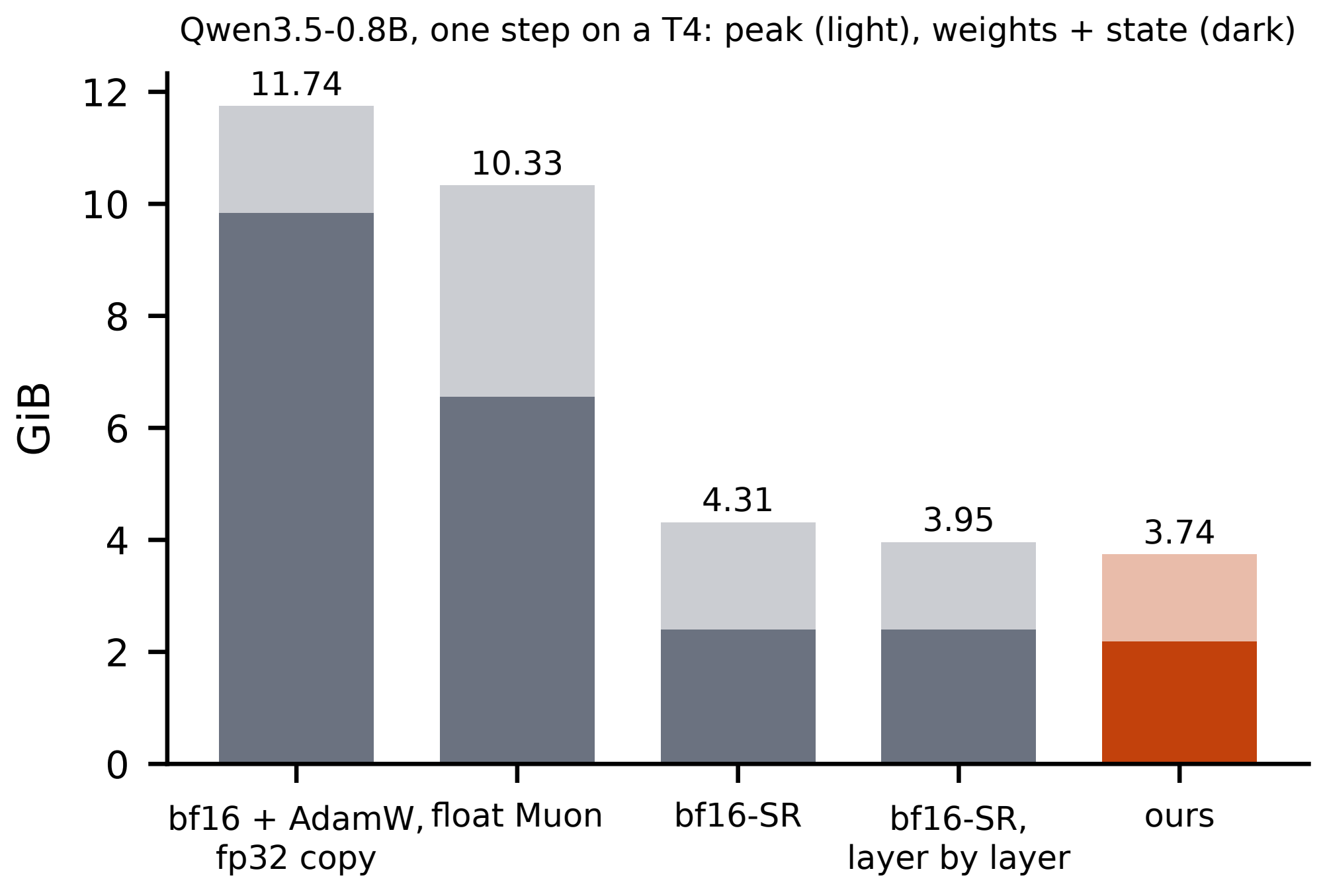
Qwen3.5-0.8B, one step on a T4: peak (light), weights + state (dark)
GiB
0
2
4
6
8
10
12
11.74
10.33
4.31
3.95
3.74
bf16 + AdamW, fp32 copy
float Muon
bf16-SR
bf16-SR, layer by layer
ours